

P2TIF: A Blockchain and Deep Learning Framework for Privacy-Preserved Threat Intelligence in Industrial IoT

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I. INTRODUCTION

Abstract—The industrial Internet of Things (IIoT) is a fast-growing network of Internet-connected sensing and actuating devices aimed to enhance manufacturing and industrial operations. This interconnection generates a high volume of data over the IIoT network and raises serious security (e.g., the rapid evolution of hacking techniques), privacy (e.g., adversaries performing data poisoning and inference attacks), and scalability issues. To mitigate the aforementioned challenges, this article presents, a new privacy-preserved threat intelligence framework (P2TIF) to protect confidential information and to identify cyber-threats in IIoT environments. There are two major elements in the proposed P2TIF framework. First, a scalable blockchain module that enables secure communication of IIoT data and prevents data poisoning attacks. Second, a deep learning module that transforms actual data into a new format and protects data from inference attacks using a deep variational autoencoder (DVAE) technique. The encoded data are then employed by a threat detection system using attention-based deep gated recurrent neural network (A-DGRNN) to recognize malicious patterns in IIoT environments. The proposed framework is validated using two different network data sources, i.e., ToN-IIoT and IoT-Botnet. Security analysis and experimental results revealed the high efficiency and scalability of the proposed P2TIF framework.

Index Terms—Blockchain, deep learning (DL), industrial Internet of Things (IIoT), privacy-preservation, threat intelligence (TI).

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THE integration of Internet of Things (IoT) technology into traditional industrial automation and control systems (IACS) has introduced major improvements to the industrial environment, and technological architecture, establishing a new network, industrial IoT (IIoT) [1]. IIoT uses IoT devices, and computation nodes to reduce resource consumption and production costs. Furthermore, it increases productivity and profitability and eventually makes the manufacturing and industrial process more intelligent [2].

In a typical IIoT environment, smart devices and sensors rely on the internet to exchange massive amounts of data and to control signals among their components. However, connecting these devices with the Internet or public channels brings about many security and privacy concerns [3], [4]. Security challenges are linked to various network-based attacks due to the rapid development in hacking techniques [5]. Physical and cyber attacks are the most common forms of threats identified in IIoT networks. Physical attacks include the actual tampering of the physical device. Such attack includes malicious code injection, and fake node injection. Whereas cyber attacks are often conducted via malicious software or malware or by gaining access to network system components [6]. Such attack includes denial-of-service (DoS), ransomware attack, man-in-the-middle (MitM), and distributed DoS (DDoS). There are also several challenges related to privacy and integrity of data where the adversary tries to steal/obtain the sensitive, private data by performing active and passive attacks [7]. The attacker attempts to sniff (private) data from the devices in active attacks (e.g., data poisoning and inference attacks). The goal of a passive attack is to collect, access, infer, or manipulate personal information [8]. This leads to an unusual consumption of resources and further stops the legitimate communication between devices [9].

Several research studies on data privacy and security measures, particularly based on access control, authentication, encryption, firewalls, and Intrusion Detection or Prevention Systems (IDS/IPS), have been introduced to protect data from unauthorized changes and to identify new attack families [10]. However, the complexity of monitoring physical and network components of the IIoT environment has shown various flaws by the aforementioned method due to the rising complexity

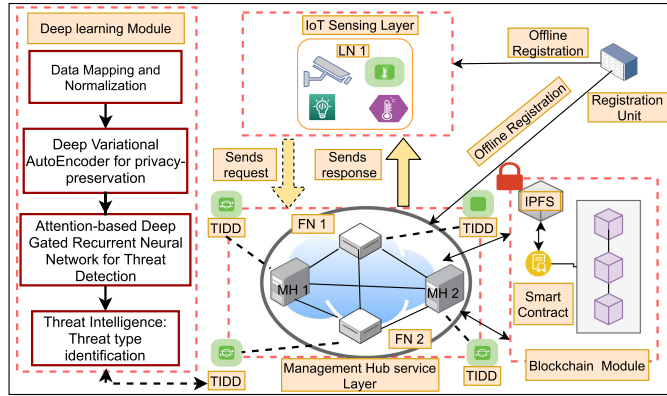


Fig. 1. Proposed P2TIF for IIoT network.

of attacks [11], [12]. Therefore, to ensure the privacy of IIoT data and security of IIoT networks, two new paradigms, i.e., blockchain and threat intelligence (TI)-based on deep learning (DL) technique can be integrated into IIoT environments.

Blockchain is a decentralized distributed ledger that has the potential to alleviate data poisoning attacks and can make the entire IIoT environment secure, robust, and cost-effective due to its cryptographically produced data blocks [13], [14]. A blockchain-enabled IIoT system can support immutability, transparency, and excellent security against a variety of aforementioned threats [15], [16]. In this article, we have designed a private blockchain, as IIoT data is highly sensitive and confidential. In private blockchain, incentives to miners are discouraged [17], [18]. Additionally, a distributed file system based on the interplanetary file system (IPFS) is utilized to make the entire IIoT environment scalable, given the limited storage of each blockchain node [19], [20]. Furthermore, utility models, such as IDS/IPS, are applied to legitimate data from the blockchain for accurate predictions, which will be beneficial for ensuring data integrity.

Threat intelligence (TI) is an emerging proactive defense technique that can learn and build knowledge about dynamic threats. A TI-based security mechanism aims to enhance the traditional attack detection process based on the evidence that can be used to discover malicious observations [21], [22]. However, its effectiveness remains uncertain due to the following two major limitations. First, due to the limited labeling of cyber threat sources, i.e., modeling TI is complex, and manually analyzing these sources to extract important threat information is a time-consuming and error-prone operation that requires a significant investment of resources [23]. Second, most prior research has used statistical and conventional machine learning (ML) techniques to develop intelligent TI schemes. Their model was difficult to apply against dynamic threats as they were complex, had low detection accuracy, had significant false alarm rates, and lacked generalization capabilities [24], [25]. Deep learning (DL) is an emerging technique that can learn the unstructured, diverse, and massive amounts of IIoT data and can be utilized to create adaptive TI models. It automatically extracts hidden vulnerability patterns from large-scale data without the need of human interaction [26]. The study of building DL-enabled TI models,

particularly for IIoT environments, is still in its early stages. In addition, DL techniques are developed as multilayer neural networks and can be used as a privacy-preservation approach for protecting confidential data (i.e., data can be safeguarded against inference attacks) from disclosure. DL technique can transform actual high-dimensional IIoT data into new low-dimensional encoded data by training multiple neural networks [27].

A. System Models

This section gives an overview of the network and threat models used to design the proposed framework.

1) **Network Model:** A P2TIF framework is proposed to detect threats and protect data in IIoT networks. The systematic architecture of the proposed framework consists of an IoT sensing layer, management hub service layer (MHSL), blockchain, and DL module, as discussed below and illustrated in Fig. 1.

- 1) *IoT sensing layer:* it includes radio-frequency identification readers, smart sensors, smart meters, and others. Due to limited resources, and computing power, these devices can store and process part of data on the blockchain and are called as *lightweight nodes* (LNs).
- 2) *The MHSL:* it consists of powerful nodes and we divide them into two nodes, i.e., *Full Nodes* (FNs) that include edge-computing servers, industrial computer and data analysis server used for mining, and *management hub* (MH), such as base station used as verifier nodes.
- 3) *Blockchain Module:* it incorporates a private blockchain, smart contract, and IPFS; concisely, a distributed file storage containing the entire transactions.
- 4) *DL Module:* it comprises a deep variational autoencoder (DVAE) technique to transform actual data into a new format and attention-based deep gated recurrent neural network (A-DGRNN)-based threat detection system to recognize malicious patterns in IIoT environment.

The proposed P2TIF framework mainly consists of LNs and FNs. The LNs have resource constraints and, therefore, can send the data to FNs available at MHSL. The FNs can assist LNs to search transactions, and can be used for mining, and adding a new block in the blockchain network. The proposed blockchain module is first used to register MH, MHSL, LNs, and FNs using trusted registration unit (RU) and then, validates the data transaction in the network using the proposed smart contract-based enhanced proof of work (ePoW). Further, we have introduced IPFS, which is distributed, content-addressable storage system used to store transactions with high integrity and without duplication [9]. In P2TIF, complete IIoT data are stored in an IPFS storage system, and the generated hash string is stored in the blockchain. In this way, P2TIF facilitates large-scale IIoT transactions with scalability. The authenticated data from the blockchain module is used by the proposed DL module to transform and detect threats. Both modules are deployed on the large distributed network or individual hosts that communicate with each other in MHSL and further coordinate to detect attacks.

2) **Threat Model:** In the proposed framework we employ the widely used “Dolev–Yao (DY) threat model,” which assumes

that all network interacting entities communicate through unsecured (public) channels [28]. An attacker, say A , has the ability to not only read the communications between them, but also to change, delete, or insert false information into the transmitted messages. In addition to the DY model, the “Canetti and Krawczyk’s model (CK-adversary model)” is also used. In CK-adversary model, A can compromise secret credentials by capturing the state of sessions. Due to this, he/she can compromise previously produced secret keys between network interacting entities by stealing session states, session keys, and short-term secrets. Furthermore, the LNs, FNs, and MH are not treated as fully trusted nodes and the RUs are considered as fully trusted entities in the IIoT environment.

B. Key Contributions

To address the abovementioned challenges, we design and implement a blockchain and DL framework hereafter referred to as P2TIF. The key contributions of this article are as follows.

- 1) A new privacy-preserved TI framework to securely transmit industrial data in IIoT network is designed by integrating blockchain and DL.
- 2) A novel blockchain module is developed to prevent data from poisoning attacks that involves three phases: i) system initialization, ii) registration and identity creation of IIoT communicating components, iii) a smart contract-based consensus mechanism to validate data records, and an IPFS-based distributed storage system to support large-scale IIoT data and enhance the scalability of the network.
- 3) A DL module is designed that has two components. The latter includes a DVAE to extract new intelligence automatically in an unsupervised manner. It also solves privacy issues by converting the actual data into a new shape or format and preventing data from inference attacks. Next, a TI scheme using A-DGRNN is designed to recognize the malicious patterns within IIoT networks.

The rest of this article is organized as follows. Section II discusses existing related work. Section III presents functional components of P2TIF framework. Experimental results are discussed in Section IV. An indepth security analysis is presented in Section V. Finally, Section VI concludes this article.

II. RELATED WORK

Several studies particularly to improve the data confidentiality and integrity in IIoT networks using blockchain and TI techniques are carried out. For instance, Singh *et al.* [15] surveyed the benefits of integrating cyber-physical systems (CPSs) with blockchain in terms of security, privacy, data/service sharing, atomicity, fault tolerance, and trust. Aggarwal *et al.* [16] surveyed the integration of smart contracts with AI in an IoT environment. Ferrag *et al.* [17] also reviewed existing blockchain protocols, and models used in IoT and its applications for maintaining privacy, security, and trust. Various research studies have been conducted to demonstrate data privacy along with threat detection in IoT and CPSs [6], [7], [8], [14]. Al-Hawawreh *et al.* [21] proposed a DL-based threat detection scheme. This

TABLE I
SUMMARY OF EXISTING WORKS

Authors	Year	A	B	C	D	E	F	G	H	I	J
Moustafa <i>et al.</i> [23]	2018	✓	✓	×	×	×	×	×	×	×	×
Muna <i>et al.</i> [5]	2018	✓	✓	×	×	×	×	×	×	×	×
Keshk <i>et al.</i> [7]	2018	✓	✓	✓	×	×	×	×	×	×	×
Yu <i>et al.</i> [2]	2019	✓	✓	×	×	×	×	×	×	×	×
Keshk <i>et al.</i> [6]	2019	✓	✓	✓	×	×	×	×	×	×	×
Al-Hawawreh <i>et al.</i> [21]	2020	✓	✓	×	×	×	×	×	×	×	×
Alkadi <i>et al.</i> [14]	2020	✓	✓	✓	×	×	✓	✓	✓	✓	✓
Keshk <i>et al.</i> [8]	2020	✓	✓	✓	×	×	✓	✓	✓	✓	✓
Ullah <i>et al.</i> [29]	2020	✓	✓	×	×	×	×	×	×	×	×
Alsaedi <i>et al.</i> [30]	2020	✓	✓	×	×	×	×	×	×	×	×
Hosen <i>et al.</i> [1]	2021	✓	✓	✓	×	×	×	×	×	×	×
Proposed (P2TIF)	2022	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

A: Threat detection; B: Security; C: Privacy; D: Scalability; E: Off-chain; F: Ledger distribution; G: Verifiability; H: Decentralized; I: Nonrepudiation; J: Transparency.

model used a deep sparse autoencoder for feature extraction and GRNN for TI-attack type identification. Booi *et al.* [10] proposed an intelligent IDS in an industrial environment. Experiments were conducted using three different classifiers: gradient boosting machine, multilayer perceptron, and random forest (RF). The proposed work was evaluated using the ToN-IoT dataset and outperformed using RF with 98.07% accuracy. Nguyen *et al.* [11] designed a graph-enabled method to detect attacks. In this model, the PSI-graph technique was used to extract features, using which a convolutional neural network (CNN)-based IDS was designed. The performance was evaluated using the IoT-Botnet dataset and the model obtained 97.8% accuracy. Gad *et al.* [12] evaluated the effectiveness of various ML-based classifiers using ToN-IoT dataset. This model used the Chi^2 technique to select features and SMOTE method to solve the class imbalance problem. The model outperformed using XGBoost and obtained 98.0% accuracy. Moustafa *et al.* [23] presented a TI approach using beta mixture-hidden Markov models in Industry 4.0. Table I summarizes the existing approaches related to the usage of blockchain and threat detection.

III. PROPOSED P2TIF FRAMEWORK

In this section, we describe the functional components used to design the proposed framework.

A. Blockchain Module

The blockchain module has three different phases, namely, a system initialization phase, a registration phase, and a validation and block creation phase. The process of each phase is discussed in detail in the upcoming sections.

1) **System Initialization:** In the phase of system initialization, the trusted RU selects various parameters as follows.

The RU first picks the nonsingular elliptic curve with form $E_p(s, t) : m^2 = n^3 + sn + t \pmod{p}$ on the Galois (finite) field $G_f(p)$, where p denotes large prime and $s, t \in Z_p^* = \{1, 2, 3, \dots, p-1\}$ are the two different constant terms such that $4s^3 + 27t^2 \not\equiv 0 \pmod{p}$ [31]. This process holds the infinity points or zero points. The RU selects the base point $L \in E_p(s, t)$ with order of $a0$ as large as point p . The RU also selects collision free unique cryptographic hash function $h_f(.)$ (for example, we use secure hash algorithms (SHA-512) as hash function). Moreover, the RU selects its own identity ID_{RU} ,

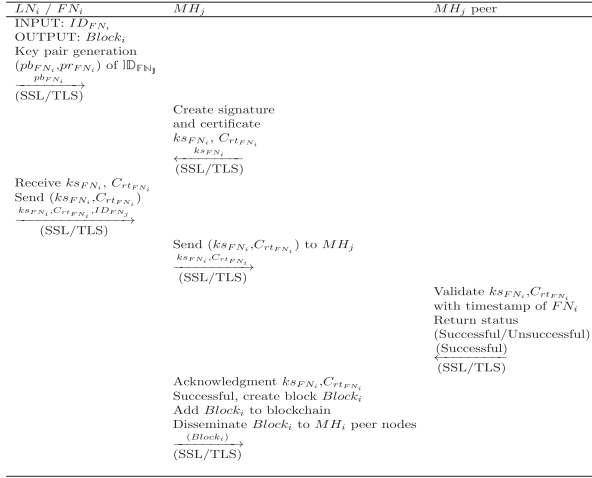


Fig. 2. Process of validation and block creation.

private key (master key) $pr_{RU} \in Z_p^*$, and computes the public key $pb_{RU} = pr_{RU} * L$. The RU keeps the secret to private key whereas RU share the parameters $\{E_p(s, t), L, h_f(\cdot), pb_{RU}\}$ as public.

If $P = (n_P, m_P)$ and $Q = (n_Q, m_Q) \in E_p(s, t)$ is two elliptic point on the curve, $R = (n_R, m_R) = P + Q$ denotes point addition on elliptic curve, which is evaluated as follows: $n_R = (\omega^2 - n_P - n_Q) \pmod{p}$; $m_R = (\omega(n_P - n_R) - m_P) \pmod{p}$; where $\omega = \left\{ \frac{(m_Q - m_P)}{(n_Q - n_P)} \right\}$, if $P \neq -Q$, $\omega = \left\{ \frac{(3n_P^2 + u)}{(2m_P)} \right\}$, if $P = Q$. The scalar point on elliptic curve is denoted as $k * L$, where $k \in Z_p^* = \{1, 2, 3, \dots, p-1\}$. This operation is further evaluated using the repetition of point doubling and addition.

2) Registration: During this phase, the trusted RU registers the MH_j securely in offline mode. In addition, the registration of MHS (fog service providers), i.e., $MHSL_j$ is done with associated FN_i and light node LN_i . The FN_i and LN_i registration node is also done in offline mode with the help of MH_j . The overall process of registration is discussed as follows.

a) **MH_j registration:** the following steps are involved with MH_j registration by the RU.

Step 1: RU selects a unique identity ID_{MH_j} for an individual MH_j , picks a random private key $pr_{MH_j} \in Z_p^*$ and finds the corresponding public key $pb_{MH_j} = pr_{MH_j} * L$, $k * L = L + L + L + \dots + L$ (k times) is the multiplication of scalar points on curve, and $k \in Z_p^*$. The RU creates a certificate for individual MH_j as $Crt_{MH_j} = pr_{MH_j} + h_f(ID_{MH_j} || ID_{RU} || pb_{RU} || pb_{MH_j} || TS_{MH_j}) * pr_{RU} \pmod{p}$, where TS_{MH_j} denotes the timestamp of the MH_j , and $*$ denotes the multiplication in Z_p^* . Once the process completes RU deletes the pr_{MH_j} from the database for security reason.

Step 2: Next, before the complete deployment with MH_j , the RU upload these information, such as $\{ID_{MH_j}, ID_{RU}, pb_{RU}, pb_{MH_j}, Crt_{MH_j}, E_p(s, t), h_f(\cdot), L\}$.

Step 3: Finally, MH_j selects a random master key $mk_{MH_j} \in Z_p^*$ and computes the public key $PB_{MH_j} = mk_{MH_j} * L$. After computation, RU publishes $\{PB_{MH_j}, pb_{RU}, pb_{MH_j}, E_p(s, t),$

$h_f(\cdot), L\}$ as public. Moreover, the MH_j has the credential of $\{PB_{MH_j}, ID_{RU}, ID_{MH_j}, Crt_{MH_j}, mk_{MH_j}, pb_{RU}, pb_{MH_j}, E_p(s, t), h_f(\cdot), L\}$.

b) **$MHSL_j$ registration:** the registration of each $MHSL_j$ is done by the MH_j using the following steps.

Step 1: The MH_j chooses a random identity ID_{MHSL_j} and generates a pseudoidentity $SID_{MHSL_j} = h_f(ID_{MHSL_j} || TS_{MHSL_j} || mk_{MH_j})$, where TS_{MHSL_j} is the timestamp for registration of $MHSL_j$. MH_j then selects a random private key $pr_{MHSL_j} \in Z_p^*$ and computes the public key $pb_{MHSL_j} = pr_{MHSL_j} * L$. Besides, MH_j creates a certificate for $MHSL_j$ as $Crt_{MHSL_j} = pr_{MHSL_j} + h_f(SID_{MHSL_j} || ID_{MH_j} || pb_{MH_j} || pb_{MHSL_j}) * mk_{MH_j} \pmod{p}$.

Step 2: The MH_j stores SID_{MHSL_j} and Crt_{MHSL_j} in its database with respective $MHSL_j$, and share pb_{MHSL_j} as public and deletes the ID_{MHSL_j} and pr_{MHSL_j} for security purpose. Further, $MHSL_j$ picks a private key $PR_{MHSL_j} \in Z_p^*$ and computes public key for same $PB_{MHSL_j} = PR_{MHSL_j} * L$.

Step 3: Finally, the MH_j preloads the credentials $\{SID_{MHSL_j}, ID_{MH_j}, Crt_{MHSL_j}, pb_{MH_j}, pb_{MHSL_j}, (PR_{MHSL_j}, PB_{MHSL_j}), PB_{MH_j}, E_p(s, t), h_f(\cdot), L\}$ in $MHSL_j$ before its deployment. In addition, MH_j keeps information of each $MHSL_j$ with $\{PB_{MHSL_j}, pb_{MHSL_j}\}$ and shares it public.

c) **LN_i and FN_i registration:** prior the deployment of IIoT nodes, i.e., (FN_i, LN_i) , MH_j registers them with the following steps.

Step 1: MH_j selects the unique identity ID_{FN_i} and computes pseudoidentity of the same $SID_{FN_i} = h_f(ID_{FN_i} || ID_{MH_j} || mk_{MH_j} || TS_{FN_i})$ for each sensor node FN_i , where TS_{FN_i} is the timestamp of registration.

Step 2: MH_j creates the private key $pr_{FN_i} \in Z_p^*$, and computes public key of same $pb_{FN_i} = pr_{FN_i} * L$, and also generates signature of nonrepudiation $ks_{FN_i} \in Z_p^*$ and its public signature $pbks_{FN_i} = ks_{FN_i} * L$, for each FN_i , whereas, for LN_i only private signature ks_{LN_i} is created.

Step 3: The MH_j creates certificates for each LN_i and FN_i as Crt_{FN_i}, Crt_{LN_i} by using $pr_{FN_i} + h_f(ID_{FN_i} || pb_{MH_j} || pb_{MHSL_j} || pb_{FN_i}) * mk_{MH_j} \pmod{p}$. After this process MH_j deletes the information ID_{FN_i} and pr_{FN_i} and stores the FN_i credentials $\{SID_{FN_i}, Crt_{FN_i}, pb_{FN_i}, (ks_{FN_i}, pbks_{FN_i}), PB_{MH_j}, E_p(s, t), h_f(\cdot), L\}$ before its deployment.

3) Validation and Block Creation Phase: Both nodes FN_i and LN_i are registered successfully by MH_j , the process of joining blockchain network takes place. The detailed steps are discussed as follows for joining the blockchain network. The ePoW is discussed in [26]. Fig. 2 shows the validation and block creation process using the following steps.

Step 1: At initial stage MH_j creates the key pair namely pb_{FN_i} and pr_{FN_i} as well as pb_{LN_i} and pr_{LN_i} for both types of nodes, i.e., FN_i and LN_i . The pb_{LN_i} and pb_{FN_i} denote as public key whereas pr_{LN_i} and pr_{FN_i} denote private key.

Step 2: Next, certificate for both the nodes is created by MH_j and also signature gets generated by FN_i and LN_i for validation of both types of nodes.

Step 3: The MH_j validates the signature of both the nodes FN_i and LN_j . After successful verification of the signature and certificate both nodes are permitted to join the network and share the data. The credential of both the nodes like pb_{FN_i} , pb_{LN_i} , $C_{rt_{FN_i}}$, and $C_{rt_{LN_i}}$ is shared over the network.

Step 4: The MH_j peers node validate the credentials like pb_{FN_i} , pb_{LN_i} , $C_{rt_{FN_i}}$, and $C_{rt_{LN_i}}$ for both the nodes using smart contracts with timestamp, which is recorded by MH_j . Once it is verified then successful/ unsuccessful message is sent to the both types of nodes, i.e., FN_i and LN_i .

Step 5: For successful message a new block ($Block_i$) is created and added to blockchain network with credentials pb_{FN_i} , pb_{LN_i} , $C_{rt_{FN_i}}$, $C_{rt_{LN_i}}$, and timestamp. The encrypted data are stored in IPFS using the algorithm mentioned in [9].

B. DL Module

After the successful execution of the proposed ePoW consensus technique, the hash is broadcasted in a blockchain-enabled IIoT network, and data (transaction) are stored in the IPFS file system in an encrypted format. Finally, the proposed DL module is designed and applied to authenticated data using two components. The first component is used to encode data into a new format. In the second component, the encoded data are used to design threat intelligence driven detection (TIDD) scheme. These two components are explained separately as follows.

1) DVAE as privacy-preservation: In this component, DVAE is applied, which is a directed probabilistic graphical unsupervised model, and is obtained by approximating the posterior of a neural network. Let us assume, we have original dataset $D = \{\phi_i\}_{i=1}^N$, which includes ϕ features and N records. The DVAE uses the latent variable v and then characterizes the distribution of D . We assume that the latent variable v 's conditional distribution is subject to Gaussian distribution (GD) [8]. Further, the theory proves that a neural network, can generate data of any distribution, if the hidden variable v satisfies the GD [32]. This implies that, by optimizing the generated θ parameter, a new dataset $\hat{D} = \{\hat{\phi}_i\}_{i=1}^N$ is produced through v , that is very much identical to initial dataset $D = \{\phi_i\}_{i=1}^N$. This implies that we want to maximize the marginal probability $m_\theta(\phi)$.

$$m_\theta(\phi) = \int m_\theta(v) m_\theta(\phi|v) dv, \text{ with } v \sim N(0, 1). \quad (1)$$

Since the actual true posterior density $m_\theta(\phi|v)$ is intractable, the DVAE implements a recognition model $r_\theta(\phi|v)$ to estimate the undetermined true posterior $m_\theta(\phi|v)$ to solve this problem. In the case of DVAE, Kullback–Leibler (KL) divergence measures the relation between recognition model $r_\theta(\phi|v)$ and true posterior distribution $m_\theta(\phi|v)$.

$$\begin{aligned} \log m_\theta(\phi^{(i)}) &= D_{KL} \left(r_\theta(v|\phi^{(i)}) || m_\theta(v|\phi^{(i)}) \right) \\ &+ G \left(\theta, \phi; \phi^{(i)} \right). \end{aligned} \quad (2)$$

Moreover, the KL divergence is always greater than 0, $\log m_\theta(\phi^{(i)}) \geq G(\theta, \phi; \phi^{(i)})$. The variational lower bound formula, $G(\theta, \phi; \phi^{(i)})$ on the marginal probability of data point ' i ' is defined as

$$\begin{aligned} G \left(\theta, \phi; \phi^{(i)} \right) &= -D_{KL} \left(r_\theta(v|\phi^{(i)}) || m_\theta(v) \right) \\ &+ E_{q_{\phi(v|\phi^{(i)})}} \left[\log m_\theta(\phi^{(i)}|v) \right]. \end{aligned} \quad (3)$$

The variational lower bound on the marginal probability represents the entire DVAE optimization objective in order to optimize $\log m_\theta(\phi)$. In the right side of (3), the first term is equivalent to the regularization term, and a negative reconstruction error is the second term. The distribution of $m_\theta(\phi^{(i)}|v)$ is presumed to be Gaussian, so $r_\theta(v|\phi^{(i)})$ can be interpreted as probabilistic rather than binary results. Thus, $r_\theta(v|\phi^{(i)})$ is a probabilistic encoder, which includes a variational parameter ϕ , and a probabilistic decoder $m_\theta(\phi^{(i)}|v)$ with a generation parameter θ [33]. Therefore, latent distributions built could decrease the probability of differences in the initial input. Using inference attacks, breaches on generated data cannot be possible, since the proposed model have the ability to produce new observations of data using various distributions possible rather than taking actual (original) data as input [8].

2) A-DGRNN as TI: Once the encoded data $\hat{D} = \{\hat{\phi}_i\}_{i=1}^N$ are obtained, the A-DGRNN-based TIDD is designed and applied for threat detection. The A-DGRNN consists of the input, hidden, middle, and output layers. The hidden layer of each A-DGRNN includes the reset with an update gate. The reset gate captures short-term information in the time series data by integrating new data with the memory. Whereas, the update gate captures long-term dependencies by determining the amount of memory needed to save the current moment. The GRU can, therefore, collect features in the time-dependent measured data and can model the mapping between targets prior measurements and the targets current states [21]. In designing GRNN-based deep neural network, input is T_c^m at each time step c hold previous hidden state $c+1$ for input vector. In addition, in the detection process the captured historical states $T_{0:c-1}^m$ are used to obtain the stored useful features from hidden layers [21]. Then, the update gate q_c and reset gate e_c are calculated using (4) and (5), respectively; where Wt_{hq} and Wt_{he} denote the weight matrices of update and reset gate, respectively. The hidden layer h_c , middle layer m_c , and output layer T_{c+1}^t are calculated using (7), (8), and (9), respectively; where Wt_{oq} , Wt_{oe} , and Wt_{oh} denote the input vector weight matrices from T_c^m of update and reset gate including $\tilde{h}_c$, respectively. Wt_{hh} denotes the weight matrices from the hidden layer including previous time steps to $\tilde{h}_c$; and $*$ represents the element-wise multiplication. The hidden states are ignored, when $e_c \approx 0$, and only the present input is used to reset the hidden states [34].

$$q_c = \sigma(Wt_{hq}h_{c-1} + Wt_{oq}T_c^m + \mathbb{b}_q) \quad (4)$$

$$e_c = \sigma(Wt_{he}h_{c-1} + Wt_{oe}T_c^m + \mathbb{b}_e) \quad (5)$$

$$\tilde{h}_c = \tanh(e_c^*Wt_{hh}h_{c-1} + Wt_{oh}T_c^m + \mathbb{b}_h) \quad (6)$$

$$h_c = (1 - q_c)^*h_{c-1} + q_c^*\tilde{h}_c \quad (7)$$

$$m_c = \tanh(Wt_{hm}h_c + \mathbb{b}_m) \quad (8)$$

$$T_{c+1}^t = \tanh(Wt_{mx}m_c + \mathbb{b}_x). \quad (9)$$

This process can help to forget the irrelevant details for the future from the present input. Wt_{hm} and Wt_{mx} are fully connected weight matrices corresponding to the middle layer and the output layer; $\sigma(\cdot)$ and $\tanh(\cdot)$ are sigmoid and hyperbolic tangent functions, respectively, and $\mathbb{b}$ represents a bias in all the above-mentioned equations. This approach can be hampered by its equal weighting of spatial and temporal features, which are not equally helpful and predictive if feature differences are ignored. As a result, we enhance our model by separating the relevance of spatial and temporal features by adding an attention mechanism to the A-DGRNN output layer T_{c+1}^t . First, the A-DGRNN output layer T_{c+1}^t is utilized to derive its implicit representation β_t via a nonlinear transformation, which may be represented as

$$\beta_t = \tanh(Wt_w T_{c+1}^t + \mathbb{b}_w). \quad (10)$$

We next use the normalized significance weight coefficient $\hat{A}_t$ to calculate the relevance of packet vectors based on the similarity representation β_t with a context vector β_w . Over β_t , Wt_w is a random initialization matrix that can concentrate on crucial information. For the coarse-grained characteristics mentioned above, the weight coefficient is [35]

$$\hat{A}_t = \frac{\exp(\beta_t^T \beta_w)}{\sum \exp(\beta_t^T \beta_w)}. \quad (11)$$

Finally, the weighted sum of T_{c+1}^t based on $\hat{A}_t$ may be used to compute the fine-grained feature ϖ_s and can be written as

$$\varpi_s = \sum \hat{A}_t T_{c+1}^t. \quad (12)$$

With a *softmax function*, the fine-grained feature vector ϖ_s created by the attention mechanism are employed for threat type identification, which can be stated as

$$y = \text{softmax}(Wt_h \varpi_s + \mathbb{b}_h) \quad (13)$$

where Wt_h is the classifier weight matrix, which can transfer ϖ_s to a new vector of length y . The number of network traffic categories is given by y .

IV. SECURITY ANALYSIS

This section discusses the security analysis of the proposed framework.

Authentication of Message: The MH_j is capable of validating the FN_i and LN_i identity ID_{FN_i} using its certificate C_{rtFN_i} and C_{rtLN_i} by using $pr_{FN_i} + h_f(ID_{FN_i} || pb_{MH_j} || pb_{MHSL_j} || pb_{FN_i}) * mk_{MH_j} \pmod{p}$. Thus, no adversary is capable of forging the valid certificate and can receive the message from FN_i and LN_i within the actual timestamp.

Privacy-Preservation: The FN_i nodes sent the message by integrating certificate C_{rtFN_i} and C_{rtLN_i} by using $pr_{FN_i} + h_f(ID_{FN_i} || pb_{MH_j} || pb_{MHSL_j} || pb_{FN_i}) * mk_{MH_j} \pmod{p}$. Further, pseudoidentity of the FN_i and LN_i gets computed using $SID_{FN_i} = h_f(ID_{FN_i} || ID_{MH_j} || mk_{MH_j} || TS_{FN_i})$ for

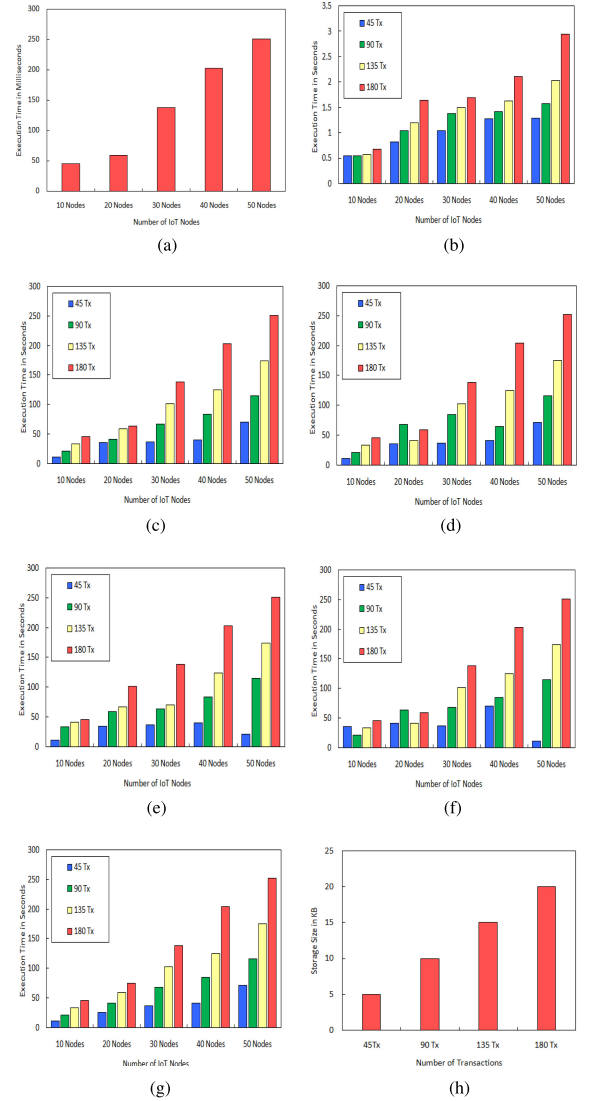


Fig. 3. Results obtained from blockchain module. (a) Registration time analysis for IoT nodes. (b) Time analysis for actual data upload into IPFS storage. (c) Time analysis of block mining for varying sizes of Tx. (d) Time analysis of block creation for varying sizes of Tx. (e) Time analysis of block access time for varying sizes of Tx. (f) Time analysis of contract deployment for varying size of Tx. (g) Time analysis of nonrepudiation for varying sizes of Tx. (h) Analysis of off-chain storage consumption (in KB) for varying size of Tx on IPFS.

each sensor node FN_i , where TS_{FN_i} denotes timestamp of FN_i registration. Thus, to get the real identity these computations are required, hence this process ensures privacy-preservation.

Replay Attack: The message and certificate of FN_i and LN_i nodes is created within a certain time interval C_{rtFN_i} and C_{rtLN_i} by using $pr_{FN_i} + h_f(ID_{FN_i} || pb_{MH_j} || pb_{MHSL_j} || pb_{FN_i}) * mk_{MH_j} \pmod{p}$. By checking its C_{rtFN_i} , C_{rtLN_i} , and the timestamp TS_{FN_i} , the MH_j can verify the message. This step stops any unapproved FN_i from transmitting the message. As a result, this procedure eliminates a replay attack.

MitM Attack: The pseudoidentity $SID_{FN_i} = h_f(ID_{FN_i} || ID_{MH_j} || mk_{MH_j} || TS_{FN_i})$ for each sensor node FN_i , where TS_{FN_i} denotes timestamp of FN_i registration and is created by

TABLE II
ADOPTED DVAE PARAMETERS

Settings	Hyperparameters
Input Layer	20 input features
Encoder	2 hidden layers: 1) Layer 1: 64 hidden nodes, a Elu function, activity_regularizer=regularizers.l1(10e-5) 2) Layer 2: 32 hidden nodes, a Elu function, activity_regularizer=regularizers.l1(10e-5)
Decoder	2 hidden layers: 1) Layer 1: 16 hidden nodes, a Elu function 2) Layer 2: 32 hidden nodes, a Elu function
DVAE Model	loss='categorical_crossentropy', optimizer='adam', epochs=20, batch_size=50

TABLE III
ADOPTED A-DGRNN PARAMETERS

Settings	Hyperparameters
Input Layer	Encoded features resulted from the DVAE technique
Hidden layers	4 hidden layers: 1) Layer 1, Layer 2, Layer 3, Layer 4: 100 hidden nodes, a tanh function, a 0.2 dropout rate.
Output layer	10 units (1 normal and 9 attack), a softmax function
A-DGRNN Model	loss='categorical_crossentropy', optimizer='adam', learning_rate=0.001, epochs=20, batch_size=50

TABLE IV
COMPARISON OF COMMUNICATION AND COMPUTATION COST

Authors	Year	Communication Cost	Computation Cost
Harishma et al. [31]	2018	160 K + 4416 bits	9.672 ms
Singh M. et al. [19]	2020	492 bits	21.4 ms
Das et al. [18]	2021	1764 bits	6.7 ms
Proposed Model	2022	577 bits	10.8 ms

the MH_j with valid certificate of C_{rtFN_i} and C_{rtLN_i} by using $pr_{FN_i} + h_f(ID_{FN_i} || pb_{MH_j} || pb_{MHSL_j} || pb_{FN_i}) * mk_{MH_j} \pmod{p}$. As a result, this procedure eliminates MitM attacks.

Impersonation Attack: To achieve an impersonation attack, adversary needs to create a pseudoidentity and certificate of FN_i and LN_i i.e., $SID_{FN_i} = h_f(ID_{FN_i} || ID_{MH_j} || mk_{MH_j} || TS_{FN_i})$ certificate C_{rtFN_i} and C_{rtLN_i} by using $pr_{FN_i} + h_f(ID_{FN_i} || pb_{MH_j} || pb_{MHSL_j} || pb_{FN_i}) * mk_{MH_j} \pmod{p}$. As a result, this procedure eliminates the impersonation attack.

V. EXPERIMENTAL RESULTS EVALUATION

The P2TIF framework uses the Keras library of TensorFlow for developing DL methods. The adopted hyper-parameters for executing DVAE and A-DGRNN are shown in Tables II and III. Ethereum and solidity (version 6.0) have been used to build a private blockchain. As specified in [9], IPFS version 0.4.19 was configured on a Tyrone PC in the specification framework. To evaluate the performance of the proposed TI scheme, different evaluation metrics, such as accuracy (AC), precision (PR), detection rate (DR), F1, and false alarm rate (FAR) are used as discussed in [9]. The obtained results are further compared after and before applying a privacy-preserving technique. Finally, results are compared with three peer ML techniques; naive Bayes (NB), RF, and decision tree (DT) based on IoT-Botnet and ToN-IoT datasets. The IoT-Botnet contains 1 498 334 threat and 79 053 normal instances [29]. The ToN-IoT includes 300 000 normal and 161 043 threat instances [30]. Both datasets are divided into train and test sets of 70%–30% and

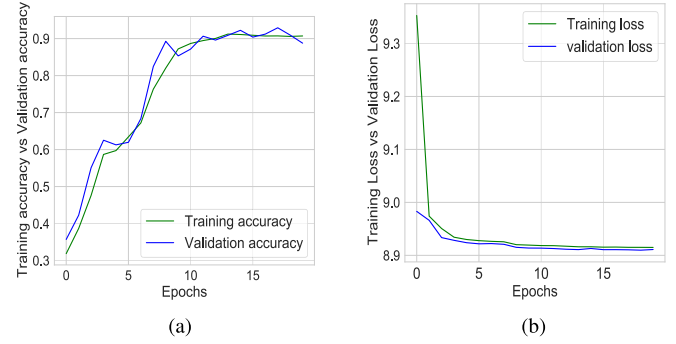


Fig. 4. The AC vs loss for DVAE technique using ToN-IoT. (a) Training vs Validation AC. (b) Training vs Validation loss.

then, data preprocessing is performed using steps mentioned in [9].

A. Analysis of Blockchain Module

In this section, we have discussed the experimental results obtained from the proposed blockchain module. First, we have registered each IoT node in the P2TIF framework to prevent malicious or unauthorized nodes from getting access to the network. Fig. 3(a) illustrates the time taken by 50 IoT nodes for registration in the network. It is seen that the time increases with IoT nodes. To make the entire framework scalable, we have stored actual data into an IPFS-based distributed file storage system. Fig. 3(b) shows the time taken by IoT nodes to store actual data into IPFS secured storage layer by varying the transactions from 45 to 180 Tx. We observe that time increases with an increase in IoT nodes and Tx. Fig. 3(c) shows the block mining time in the network by the proposed ePoW. Once the block is mined, then, the block is created in the network. Fig. 3(d) shows block creation time and Fig. 3(e) illustrates block access time. It is seen with an increase in Tx and IoT nodes, the execution time increases gradually. The smart contract deployment time in the network is shown in Fig. 3(f). The digital signature of IoT devices is created in the network to manage nonrepudiation in the network. Fig. 3(g) shows the time taken to ensure nonrepudiation by the P2TIF framework. Fig. 3(h) illustrates the storage size consumption by different Tx in the network. It is seen that with an increase in the network the storage time increases gradually. In addition, we have calculated the computation and communication costs of the proposed framework as follows.

1) Computation Cost: The computation cost is evaluated based on addition, multiplication, scalar entity, and modulus operation performed in each phase. The computation cost for FN_i is computed as follows.

Computation time for FN_i registration: the FN_i includes the random identity selection, which takes 0.1 ms. The pseudoidentity gets created with four scalar parameters including timestamp, which takes 4.1 ms. In addition, the overall process takes 3 multiplication, 1 addition, 1 modulus operation, and 1 comparisons operation. By adding, the total computation cost takes 6.6 ms. Hence, the total cost for the registration of FN_i is $4.1 + 0.1 + 6.6 = 10.8$ ms.

TABLE V
CLASS-WISE PREDICTION (%) OBTAINED FROM A-DGRNN-BASED TIDD BASED ON ToN-IoT DATASET

Dataset	Parameters	Backdoor	DDoS	DoS	Injection	MITM	Normal	Password	Ransomware	Scanning	XSS
With Original	PR	99.98	96.63	99.89	93.49	98.98	100.00	99.91	99.86	99.98	97.93
	DR	99.95	96.69	95.97	97.12	96.71	100.00	97.99	100.00	100.00	99.91
	F1	99.96	96.66	97.89	95.27	97.83	100.00	98.94	99.93	99.99	98.91
	FAR	0.000007	0.00153	0.00004	0.00303	0.00002	0.00000	0.00003	0.00006	0.000007	0.00093
With Transformed	PR	99.99	95.06	99.27	93.27	88.10	100.00	99.59	99.35	98.74	97.72
	DR	99.98	94.98	94.78	96.25	90.13	99.99	99.08	99.96	98.50	99.21
	F1	99.95	95.02	96.98	94.73	89.10	99.99	99.33	99.65	98.62	98.46
	FAR	0.00003	0.00224	0.00031	0.00312	0.00026	0.00000	0.00018	0.00029	0.00056	0.00102

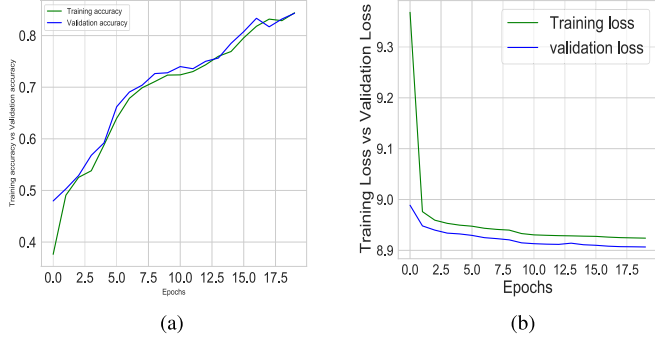


Fig. 5. The AC vs loss for DVAE technique using IoT-Botnet. (a) Training vs Validation Acc. (b) Training vs Validation loss.

TABLE VI
CLASS-WISE PREDICTION (%) OBTAINED FROM A-DGRNN-BASED TIDD BASED ON IoT-BOTNET

Dataset	Parameters	DoS	DDoS	Normal	Reconnaissance	Theft
With Original	PR	100.00	99.99	99.99	99.99	80.95
	DR	100.00	99.99	99.99	100.00	70.83
	F1	100.00	99.98	99.99	99.99	75.55
	FAR	0.0000	0.0001	0.000006	0.000004	0.00003
With Transformed	PR	85.10	97.15	94.80	99.24	02.88
	DR	94.96	80.72	99.93	99.32	18.75
	F1	89.76	88.17	97.30	99.28	05.00
	FAR	0.08404	0.01156	0.02182	0.00039	0.00138

Computation time for transaction: for the transaction process, pseudoidentity is checked, which takes 4 scalar parameters that takes 4 ms. In the blockchain 1 comparison and 1 mapping check are required, which takes a total of 6 ms. Finally, the verification of the smart contract takes 5 ms. The total computation in transaction process takes (registration time + transaction time), i.e., $10.8 + 15 = 25.8$ ms.

2) Communication Cost: Communication cost involves the key management and the process of credential sharing over the network. The proposed model communication cost is calculated as follows.

Communication time for FN_i registration: in the process of registration phase the key management process involved $\{SID_{FN_i}, C_{rt_{FN_i}}, pb_{FN_i}, (ks_{FN_i}, pbks_{FN_i}), PB_{MH_j}, E_p(s, t), h_f(\cdot), L\}$, where pseudoidentity $\{SID_{FN_i}$ takes 32 bits of size, certificate takes 224 bits of size. The public key of FN_i takes 32 bits and public key of PB_{MH_j} takes 32 bits of size. The curve point takes 1 bit of size, and the hash computation takes 256 bits of size. The total communication cost in key management takes $32 + 224 + 32 + 32 + 1 + 256 = 577$ bits. The communication cost is evaluated here for deployment over the network for FN_i .

Communication time for transaction: in the process of communication cost 32 bits of identity is created for FN_i node. The

9 bits of information is used for the status of FN_i node. The mapping of a message is created with a content-addressed hash of 48 bits. On the blockchain side communication cost of 32 bits of signature, 256 bits of hash, and 32 bits of the key generation process is evaluated. The total cost is calculated by adding all parameters takes 409 bits of size. Finally, the total communication cost is evaluated by adding FN_i registration and blockchain side computation, i.e., $577 + 409 = 986$ bits. Table IV shows the comparisons of communication cost and computation cost. It can be seen that the proposed model takes almost approximate computation cost whereas communication cost is efficient than other existing models.

B. Analysis of DL Module

Once the proposed ePoW is executed, the DL-based DVAE approach is applied to reshape IoT data into a new format. This approach prevents data inference attacks that can be used during the training phase of the model. The AC and loss results obtained from DVAE revealed that the model could efficiently learn from both IoT network datasets, as illustrated in Figs. 4 and 5. Moreover, the DVAE model has obtained about 89.99% validation AC whereas 8.81% recorded as a loss with the ToN-IoT dataset. Similarly, the model has achieved 89.90% validation AC and 8.98% loss with the IoT-Botnet dataset. The DVAE aim is not to detect these cyber-threats but to encode the data into a new dimension and further this encoded data are used to design a TIDD scheme. The A-DGRNN-based TIDD was trained and validated using the encoded data in the training phase. Based on the evaluation metrics of PR, DR, F1, and FAR, Tables V and VI show class wise overall performances based on ToN-IoT and IoT-Botnet datasets. Further, we have compared the performance before and after preserving sensitive attributes in both datasets. For both network datasets, the values of PR, DR, and F1 lie between 88.10%–100%, and FAR is close to zero.

C. Results Comparison With Baseline Techniques

Tables VII and VIII show comparison of DR between A-DGRNN-based TIDD, i.e., before and after applying privacy-preserving and other peer privacy-preserving TI techniques, i.e., RF, DT, and NB. It can be seen that our approach has outperformed other techniques, and has obtained DR of 95–100% for various IoT-based attacks in both cases (i.e., with original and transformed) with the ToN-IoT dataset. Similarly, for the IoT-Botnet dataset, DR was recorded between 80.72–100% for normal, DoS, DDoS, and reconnaissance attacks. However, for theft attacks using the original IoT-Botnet dataset DR is 70.83%

TABLE VII
COMPARISON OF MULTIVECTOR DETECTION RATE (DR) (%) WITH BASELINE APPROACHES USING ToN-IoT DATASET

Methods	Backdoor	DDoS	DoS	Injection	MITM	Normal	Password	Ransomware	Scanning	XSS
RF	99.98	90.40	91.97	93.53	0.00	100.00	97.81	99.40	95.74	85.47
DT	100.00	100.00	100.00	0.00	0.00	100.00	100.00	100.00	100.00	100.00
NB	99.22	26.80	91.70	92.96	95.11	100.00	75.32	79.98	96.91	19.02
A-DGRNN (with Original)	99.95	96.69	95.97	97.12	96.71	100.00	97.99	100.00	100.00	99.91
A-DGRNN (with Transformed)	99.98	94.98	94.78	96.25	90.13	99.99	99.08	99.96	98.50	99.21

TABLE VIII
COMPARISON OF MULTIVECTOR DETECTION RATE (DR) (%) WITH BASELINE APPROACHES USING IoT-BOTNET DATASET

Detection Method	Normal	DoS	DDoS	Reconnaissance	Theft
RF	14.95	99.96	100.00	100.00	0.00
DT	0.00	100.00	100.00	80.06	100.00
NB	74.76	97.76	99.97	81.44	92.85
A-DGRNN (with Original)	100.00	99.99	99.99	100.00	70.83
A-DGRNN (with Transformed)	94.96	80.72	99.93	99.32	18.75

TABLE IX
PERFORMANCE COMPARISON WITH EXISTING STATE-OF-THE-ART THREAT DETECTION TECHNIQUES

Authors	Year	Method	Dataset	Accuracy
Keshk et al. [7]	2018	DT-ICA	Power Data	97.43%
Keshk et al. [6]	2019	PPAD-CPS	Power System	97.27%
Nguyen et al. [11]	2020	CNN	IoT-Botnet	98.70%
Keshk et al. [8]	2020	LSTM	Power System	96.27%
Alsaedi et al. [30]	2020	CART	ToN-IoT	77.00%
Booij et al. [10]	2021	RF	ToN-IoT	98.07%
Dunn et al. [12]	2021	XGBoost	ToN-IoT	98.00%
Proposed Work	2022	P2TIF	ToN-IoT (Original)	99.45%
			ToN-IoT (Transformed)	99.88%
			IoT-Botnet (Original)	99.03%
			IoT-Botnet (Transformed)	99.29%

Terms and abbreviations: DT-ICA: Decision tree-independent component analysis, PPAD-CPS: Privacy-preserving anomaly detection cyber-physical systems, CART: Classification and regression trees, LSTM: Long- short-term memory.

and with the transformed dataset DR is 18.75%. This is due to the low instances present in theft attacks.

D. Results Comparisons With Existing Techniques

Table IX shows performance comparison in terms of accuracy with existing state-of-the-art approaches. We see that few IDS effectiveness were tested using obsolete datasets, such as power data and system [6]–[8]. Moreover, these datasets have less practical value for a modern IDS, as they lack current attack vectors. In [10], [11], [12], and [30] few recent datasets, such as IoT-Botnet and ToN-IoT were used for evaluation. We see that the proposed framework P2TIF has outperformed these state-of-the-art methods using both datasets in terms of accuracy in both scenarios, i.e., with and without privacy-preserving technique (original and transformed). The P2TIF framework has achieved high values for the abovementioned evaluation metric, this is due to the inclusion of blockchain, high performance of DL techniques, and its capability to classify time-series data, such as ToN-IoT and IoT-Botnet.

VI. CONCLUSION

In this article, we designed and implemented a new framework named P2TIF, using blockchain and DL. The proposed framework protects industrial data and efficiently identify malicious events in IIoT networks. First, the blockchain module was designed to register, verify, and validate transactions using smart contract enabled ePoW. In addition, IPFS distributed storage system was employed to make the P2TIF framework scalable and store encrypted industrial transactions. Second,

a DL module with two components was designed that used authenticated data from the blockchain module for threat identification. In the first DL component, a DVAE technique was employed to automatically extract new intelligence in an unsupervised manner. It also solved privacy issues (i.e., rescues data from inference attacks) by transforming industrial data into a new format. Next, an A-DGRNN-based TIDD was designed and evaluated in two different scenarios, i.e., before and after applying a privacy-preservation technique. The results showed that the proposed approach using original and transformed data can outperform other peer privacy-preserving TI techniques and some recent state-of-the-art works. Our future directions will include utilizing a microservice architecture to evaluate the scalability and utility of the P2TIF framework.

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